

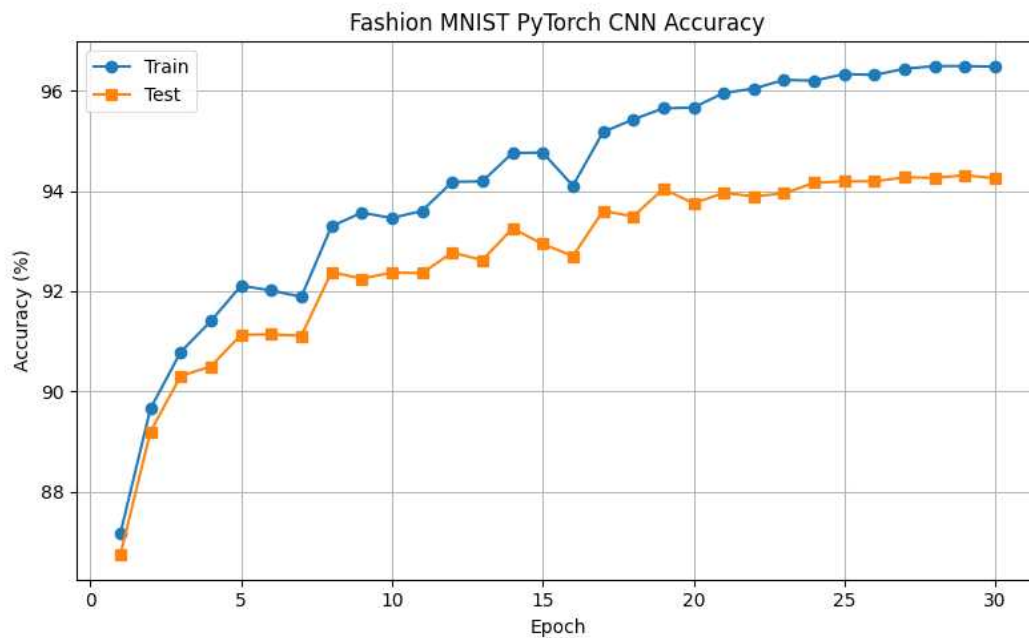
인공지능개론 미니프로젝트

ICT융합공학부 202204076 안민성

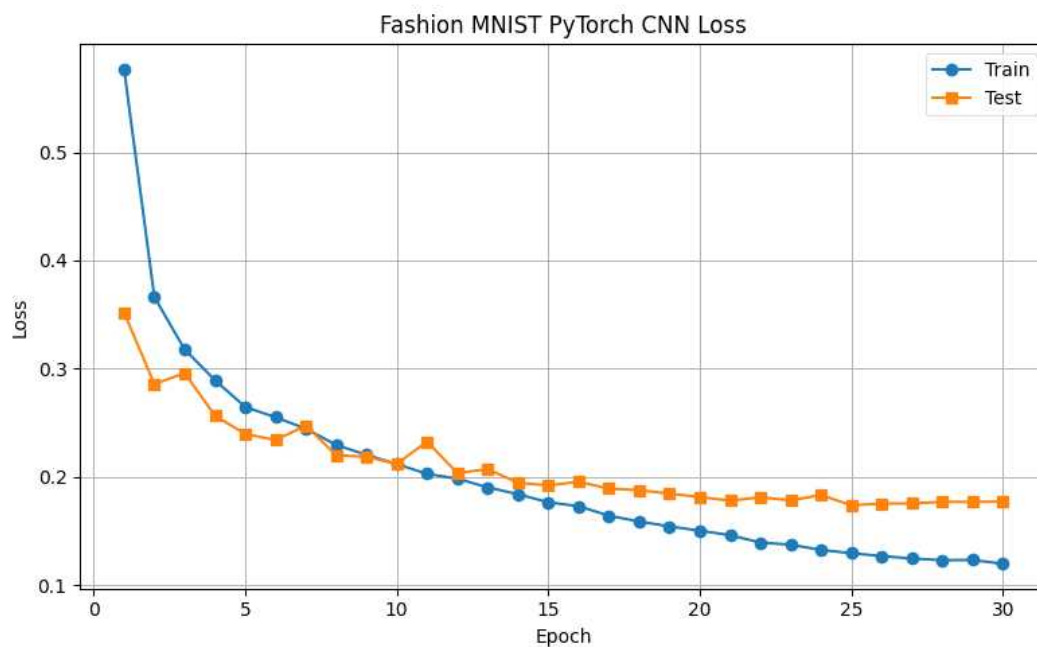
ICT융합공학부 202204091 이다균

1. pytorch

1-1)정확도 그래프



1-2)손실함수 그래프



1-3)로그

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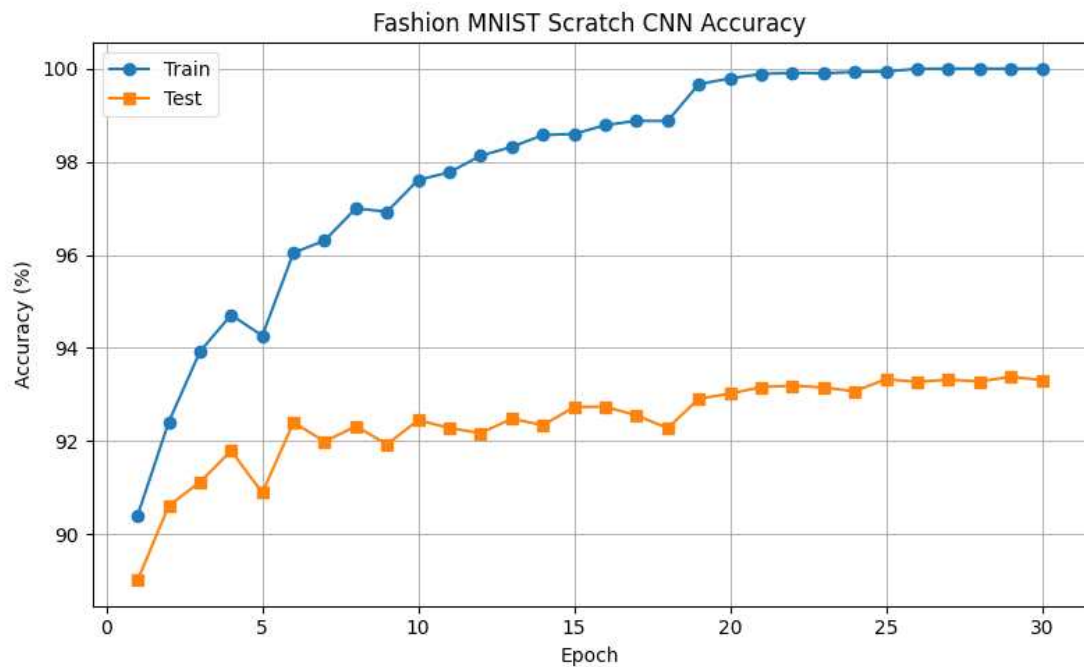
Device: cuda
Epoch [01/30] LR: 0.000997, Train Loss: 0.5769, Train Acc: 87.11%, Test Loss: 0.3519, Test Acc: 86.46%
Epoch [02/30] LR: 0.000989, Train Loss: 0.3662, Train Acc: 90.12%, Test Loss: 0.2858, Test Acc: 89.26%
Epoch [03/30] LR: 0.000976, Train Loss: 0.3177, Train Acc: 89.90%, Test Loss: 0.2960, Test Acc: 89.38%
Epoch [04/30] LR: 0.000957, Train Loss: 0.2894, Train Acc: 91.53%, Test Loss: 0.2566, Test Acc: 90.51%
Epoch [05/30] LR: 0.000934, Train Loss: 0.2647, Train Acc: 92.25%, Test Loss: 0.2397, Test Acc: 91.23%
Epoch [06/30] LR: 0.000905, Train Loss: 0.2552, Train Acc: 92.48%, Test Loss: 0.2341, Test Acc: 91.64%
Epoch [07/30] LR: 0.000873, Train Loss: 0.2446, Train Acc: 91.97%, Test Loss: 0.2475, Test Acc: 91.15%
Epoch [08/30] LR: 0.000836, Train Loss: 0.2296, Train Acc: 93.26%, Test Loss: 0.2200, Test Acc: 92.19%
Epoch [09/30] LR: 0.000796, Train Loss: 0.2204, Train Acc: 93.47%, Test Loss: 0.2186, Test Acc: 92.39%
Epoch [10/30] LR: 0.000753, Train Loss: 0.2119, Train Acc: 93.61%, Test Loss: 0.2121, Test Acc: 92.38%
Epoch [11/30] LR: 0.000706, Train Loss: 0.2029, Train Acc: 92.78%, Test Loss: 0.2327, Test Acc: 91.38%
Epoch [12/30] LR: 0.000658, Train Loss: 0.1984, Train Acc: 94.57%, Test Loss: 0.2035, Test Acc: 93.17%
Epoch [13/30] LR: 0.000608, Train Loss: 0.1904, Train Acc: 94.25%, Test Loss: 0.2072, Test Acc: 92.50%
Epoch [14/30] LR: 0.000557, Train Loss: 0.1840, Train Acc: 94.83%, Test Loss: 0.1946, Test Acc: 93.27%
Epoch [15/30] LR: 0.000505, Train Loss: 0.1766, Train Acc: 94.77%, Test Loss: 0.1922, Test Acc: 92.92%
Epoch [16/30] LR: 0.000453, Train Loss: 0.1730, Train Acc: 94.63%, Test Loss: 0.1955, Test Acc: 93.15%
Epoch [17/30] LR: 0.000402, Train Loss: 0.1642, Train Acc: 95.28%, Test Loss: 0.1893, Test Acc: 93.45%
Epoch [18/30] LR: 0.000352, Train Loss: 0.1589, Train Acc: 95.61%, Test Loss: 0.1878, Test Acc: 93.64%
Epoch [19/30] LR: 0.000304, Train Loss: 0.1543, Train Acc: 95.60%, Test Loss: 0.1846, Test Acc: 93.53%
Epoch [20/30] LR: 0.000258, Train Loss: 0.1504, Train Acc: 95.73%, Test Loss: 0.1815, Test Acc: 93.82%
Epoch [21/30] LR: 0.000214, Train Loss: 0.1462, Train Acc: 95.93%, Test Loss: 0.1781, Test Acc: 93.82%
Epoch [22/30] LR: 0.000174, Train Loss: 0.1394, Train Acc: 96.03%, Test Loss: 0.1813, Test Acc: 94.01%
Epoch [23/30] LR: 0.000137, Train Loss: 0.1374, Train Acc: 96.27%, Test Loss: 0.1784, Test Acc: 93.93%
Epoch [24/30] LR: 0.000105, Train Loss: 0.1326, Train Acc: 96.33%, Test Loss: 0.1833, Test Acc: 94.04%
Epoch [25/30] LR: 0.000076, Train Loss: 0.1298, Train Acc: 96.38%, Test Loss: 0.1739, Test Acc: 94.24%
Epoch [26/30] LR: 0.000053, Train Loss: 0.1269, Train Acc: 96.39%, Test Loss: 0.1754, Test Acc: 94.10%
Epoch [27/30] LR: 0.000034, Train Loss: 0.1246, Train Acc: 96.48%, Test Loss: 0.1756, Test Acc: 94.27%
Epoch [28/30] LR: 0.000021, Train Loss: 0.1232, Train Acc: 96.53%, Test Loss: 0.1770, Test Acc: 94.25%
Epoch [29/30] LR: 0.000013, Train Loss: 0.1233, Train Acc: 96.59%, Test Loss: 0.1770, Test Acc: 94.27%
Epoch [30/30] LR: 0.000010, Train Loss: 0.1198, Train Acc: 96.57%, Test Loss: 0.1772, Test Acc: 94.24%

Final Result
Train Accuracy: 96.57%
Test Accuracy: 94.24%

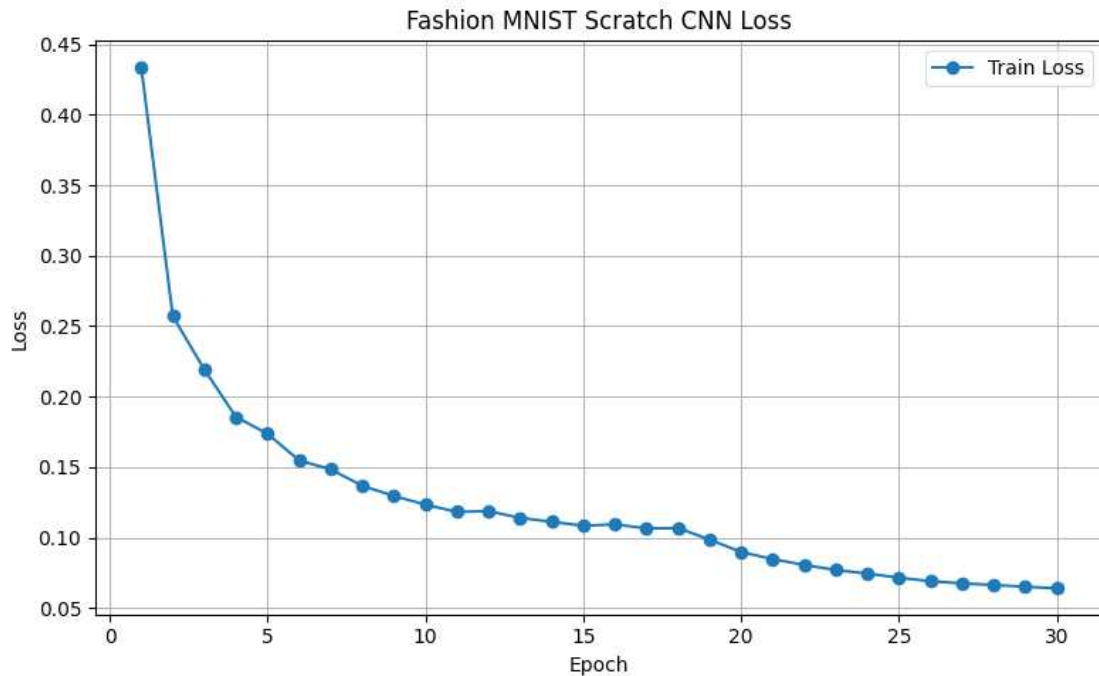
Saved files:
- logs.txt
- accuracy_graph.png
- loss_graph.png
- fashion_cnn.pth
  
```

2. scratch

2-1)정확도 그래프



2-2)손실함수 그래프



2-3)로그

```
x_train: (60000, 1, 28, 28)
t_train: (60000,)
x_test: (10000, 1, 28, 28)
t_test: (10000,)
Epoch [01/30] Iter [468/14040] LR: 0.001000, Loss: 0.4338, Train Acc: 90.40%, Test Acc: 89.01%
Epoch [02/30] Iter [936/14040] LR: 0.001000, Loss: 0.2570, Train Acc: 92.41%, Test Acc: 90.61%
Epoch [03/30] Iter [1404/14040] LR: 0.001000, Loss: 0.2192, Train Acc: 93.92%, Test Acc: 91.12%
Epoch [04/30] Iter [1872/14040] LR: 0.001000, Loss: 0.1856, Train Acc: 94.71%, Test Acc: 91.79%
Epoch [05/30] Iter [2340/14040] LR: 0.001000, Loss: 0.1736, Train Acc: 94.26%, Test Acc: 90.89%
Epoch [06/30] Iter [2808/14040] LR: 0.001000, Loss: 0.1547, Train Acc: 96.04%, Test Acc: 92.40%
Epoch [07/30] Iter [3276/14040] LR: 0.001000, Loss: 0.1484, Train Acc: 96.31%, Test Acc: 91.99%
Epoch [08/30] Iter [3744/14040] LR: 0.001000, Loss: 0.1367, Train Acc: 97.00%, Test Acc: 92.32%
Epoch [09/30] Iter [4212/14040] LR: 0.001000, Loss: 0.1295, Train Acc: 96.92%, Test Acc: 91.93%
Epoch [10/30] Iter [4680/14040] LR: 0.001000, Loss: 0.1234, Train Acc: 97.61%, Test Acc: 92.45%
Epoch [11/30] Iter [5148/14040] LR: 0.001000, Loss: 0.1182, Train Acc: 97.77%, Test Acc: 92.28%
Epoch [12/30] Iter [5616/14040] LR: 0.001000, Loss: 0.1187, Train Acc: 98.13%, Test Acc: 92.17%
Epoch [13/30] Iter [6084/14040] LR: 0.001000, Loss: 0.1140, Train Acc: 98.32%, Test Acc: 92.48%
Epoch [14/30] Iter [6552/14040] LR: 0.001000, Loss: 0.1112, Train Acc: 98.57%, Test Acc: 92.34%
Epoch [15/30] Iter [7020/14040] LR: 0.001000, Loss: 0.1084, Train Acc: 98.60%, Test Acc: 92.73%
Epoch [16/30] Iter [7488/14040] LR: 0.001000, Loss: 0.1094, Train Acc: 98.78%, Test Acc: 92.74%
Epoch [17/30] Iter [7956/14040] LR: 0.001000, Loss: 0.1065, Train Acc: 98.88%, Test Acc: 92.55%
Epoch [18/30] Iter [8424/14040] LR: 0.000300, Loss: 0.1067, Train Acc: 98.87%, Test Acc: 92.28%
Epoch [19/30] Iter [8892/14040] LR: 0.000300, Loss: 0.0985, Train Acc: 99.67%, Test Acc: 92.91%
Epoch [20/30] Iter [9360/14040] LR: 0.000300, Loss: 0.0898, Train Acc: 99.78%, Test Acc: 93.02%
Epoch [21/30] Iter [9828/14040] LR: 0.000300, Loss: 0.0847, Train Acc: 99.89%, Test Acc: 93.16%
Epoch [22/30] Iter [10296/14040] LR: 0.000300, Loss: 0.0806, Train Acc: 99.91%, Test Acc: 93.19%
Epoch [23/30] Iter [10764/14040] LR: 0.000300, Loss: 0.0770, Train Acc: 99.90%, Test Acc: 93.15%
Epoch [24/30] Iter [11232/14040] LR: 0.000300, Loss: 0.0744, Train Acc: 99.93%, Test Acc: 93.07%
Epoch [25/30] Iter [11700/14040] LR: 0.000090, Loss: 0.0716, Train Acc: 99.94%, Test Acc: 93.33%
Epoch [26/30] Iter [12168/14040] LR: 0.000090, Loss: 0.0690, Train Acc: 100.00%, Test Acc: 93.27%
Epoch [27/30] Iter [12636/14040] LR: 0.000090, Loss: 0.0675, Train Acc: 100.00%, Test Acc: 93.32%
Epoch [28/30] Iter [13104/14040] LR: 0.000090, Loss: 0.0663, Train Acc: 100.00%, Test Acc: 93.28%
Epoch [29/30] Iter [13572/14040] LR: 0.000090, Loss: 0.0651, Train Acc: 100.00%, Test Acc: 93.38%
Epoch [30/30] Iter [14040/14040] LR: 0.000090, Loss: 0.0640, Train Acc: 100.00%, Test Acc: 93.31%
Calculating final accuracy...
```

```
Final Result
Train Accuracy: 100.00%
Test Accuracy: 93.31%
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Saved files:
- logs.txt
- accuracy_graph.png
- loss_graph.png
- scratch_params.pkl
```